

1. The Setup: Normalizing (Scaling Down) House price prediction

Before the math started, we turned big numbers into small ones (0 to 1).

- Input: 50 sqm $\rightarrow$ 0.0
- Input: 80 sqm $\rightarrow$ 1.0
- Target Price: 150k $\rightarrow$ 0.0
- Target Price: 240k $\rightarrow$ 1.0

Why? It makes the math easier and prevents the "Exploding Numbers" error we saw earlier.

2. The Prediction Formula

The model makes a guess using this simple line equation: $\text{Guess} = (\text{Weight} \times \text{Input}) + \text{Bias}$

- Weight (w): How much the price changes when size changes. (Started at 0.4972).
- Bias (b): A fixed starting price. (Started at 0.0).
- Input (x): The house size (e.g., 0.33 for a 60sqm house).

Example Calculation (First Try):

- House Size: 60 sqm (Normalized: 0.33)
 - Current Weight: 0.4972
 - Current Bias: 0.0
 - Calculation: $(0.4972 \times 0.33) + 0.0 = \mathbf{0.164}$
 - Real Price (Target): 0.33 (normalized 180k)
 - Mistake: The model guessed 0.164, but the real answer is 0.33. It was too low.
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3. The Error (The "Oops" Moment)

The model calculates how wrong it was: $\text{Error} = \text{Target} - \text{Guess}$
 $\text{Error} = 0.33 - 0.164 = \mathbf{0.166}$

This number (0.166) tells the model: "You are 0.166 units too low!"

4. The Update (The "Learning" Step)

This is the most important part. The model adjusts its Weight and Bias to get closer to the answer next time.

The rule is:
$$\text{New Weight} = \text{Old Weight} + (\text{Learning Rate} \times \text{Error} \times \text{Input})$$
$$\text{New Bias} = \text{Old Bias} + (\text{Learning Rate} \times \text{Error})$$

The Math with your numbers:

- Learning Rate: 0.1 (The step size)
- Error: 0.166
- Input: 0.33

Calculating the Change for Weight:

1. Multiply Error by Input: $0.166 \times 0.33 = 0.055$
2. Multiply by Learning Rate: $0.055 \times 0.1 = \mathbf{0.0055}$
3. Add to Old Weight: $0.4972 + 0.0055 = \mathbf{0.5027}$

Calculating the Change for Bias:

1. Multiply Error by Learning Rate: $0.166 \times 0.1 = \mathbf{0.0166}$
2. Add to Old Bias: $0.0 + 0.0166 = \mathbf{0.0166}$

Result: The model now has a new, better weight (0.5027) and a new bias (0.0166).

- Next time it sees a 60sqm house, it will guess slightly higher, getting closer to the real price.

5. The Loop (Repeating 2000 Times)

The code does this for every house in your list (50, 60, 70, 80 sqm), then starts over from the beginning.

- Epoch 1: It makes small mistakes.
- Epoch 500: The mistakes are tiny. The weight has slowly moved from 0.49 to 0.99.
- Epoch 2000: The weight is almost exactly 1.0.

Why did it end at 1.0?

Because the real world data was perfectly linear:

- If Size goes up by 1 unit, Price goes up by 1 unit.
- The math naturally pushes the Weight to 1.0 to match that ratio.
- The Bias stays 0.0 because when Size is 0, Price is 0.